S Murugan

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Summary

Computer Science undergraduate with strong foundations in Java, data structures, algorithms, and object-oriented programming. Experienced in building backend and full-stack applications using REST APIs, databases, and modern web technologies. Passionate about software engineering, problem-solving and developing scalable user-focused solutions.

Education

NSS College of Engineering — Palakkad, India
Bachelor of Technology in Computer Science and Engineering | CGPA: 8.89 / 10 — *Aug 2023 – Present*

APHSS Adichanalloor — Kollam, India
Higher Secondary (PCMB) | Percentage: 98% — *2022*

Experience

Generative AI Intern — Jul 2025 – Feb 2026
Akiyam Solutions Pvt. Limited — *Remote*

- Contributed to backend development for an HRMS platform by implementing RBAC authorization, task orchestration, and REST API features as part of a 9-engineer team.
- Developed a hybrid retrieval pipeline combining SPLADE, TF-IDF, and dense embeddings for large-scale document retrieval and ranking.
- Automated a text-to-video generation workflow using the Wan 2.1 model and containerized backend services with Docker to streamline development and deployment.

Projects

DICE-Flow – Disease Gene Discovery Pipeline | *Python, C++ Graph Engine, FastAPI, React, Docker*

- Academic group mini project using a C++ graph engine as the base for SSSP and centrality analysis on weighted protein-interaction graphs.
- Included biomarker ranking with Brandes' betweenness, eigenvector centrality, virtual gene-knockout simulation, FastAPI services, and a React/Cytoscape.js dashboard.

Learnix.ai | GitHub | *Python, Flask, LangChain, ChromaDB, Supabase, Next.js*

- Engineered a RAG layer over ChromaDB and a paper-ingestion pipeline using pypdf, Tesseract OCR fallback, and Llama 3.1-8B via Groq to parse CBSE/KTU papers into structured JSON.
- Built an 18-endpoint Flask REST API on Supabase PostgreSQL with admin-scoped Row-Level Security, integrated with a Next.js frontend and a Monte Carlo pass-probability simulator running 10,000 iterations per query.

Expenso | GitHub | *Flutter, Supabase, Gemini 2.5 Flash, VAPI, WebRTC*

- Built a voice-based personal finance prototype converting natural speech into multi-step transactions; reduced logging time from over 10 seconds to under 2 seconds.
- Connected LLM function-calling flows with VAPI/WebRTC and Supabase; tested with 50+ active users and 500+ logged transactions.

Technical Skills

Languages: Java, Python, JavaScript, SQL
Frameworks: FastAPI, Flask, React, Node.js, REST APIs, HTML, CSS
Databases: MySQL, MongoDB, Supabase
Developer Tools: Git, GitHub, Docker, Linux, CI/CD
CS Fundamentals: DSA, OOP, DBMS, Operating System, Computer Networks
AI / ML: LLMs, RAG, LangChain, OCR

Achievements

1st Prize | HackNova, NSS College of Engineering — Sep 2025
2nd Runner-Up | Hack-4-Mini 2.0 - CodeOholics National Hackathon — Jun 2025
25th Global Rank | RedFox CTF 2026 — Mar 2026
29th Global Rank | PsychCTF 2026 — Mar 2026
IBM SkillsBuild Certification | Getting Started With Artificial Intelligence